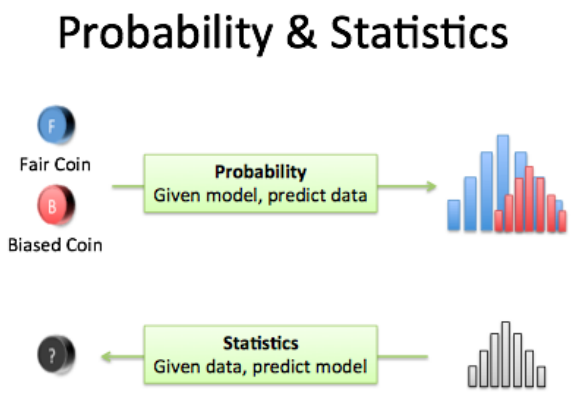
Chapter 5: Statistics

Math 217: Probability & Statistics

Distiction btw Prob. and Stat.

- **Probability and statistics are closely related and each depends on the other in a number of different ways.**
- A knowledge or no knowledge about the distribution
  - For Instance: How to check the fairness of the coin.



Type of Statistics

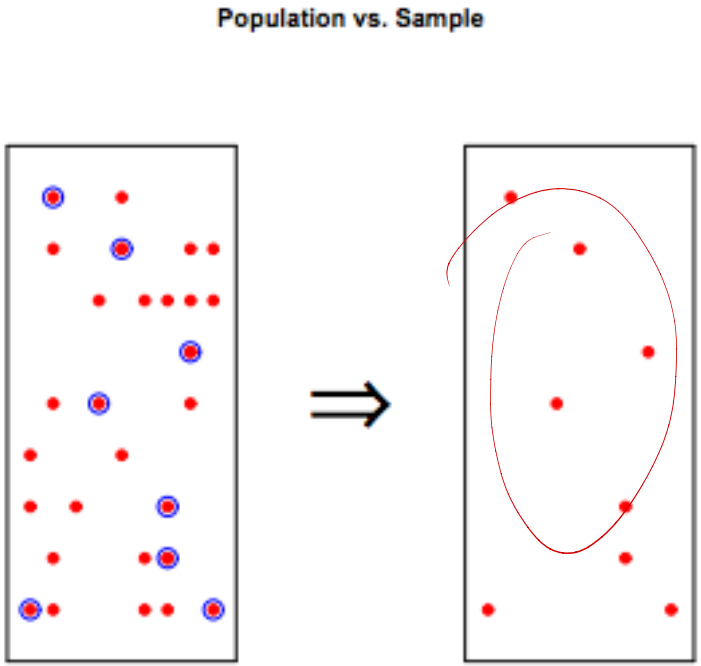
- Two general subjects of area
  - Descriptive statistics
  - Inferential statistics

Key terms used in the field of statistics

- A **population** includes all of the element from a set of data. It is a huge data which we have not completely examined. The population size is shown as upper case N
- A **sample** consists of one or more observations from the population., That portion of the population that is available, or to be made available, for analysis.

- We will learn about probability samples and how they provide assurance that a sample is indeed representative. The sample size is shown as lower case n

Ex: If your company manufactures one million laptops, they might take a sample of say, 500, of them to test quality. The population N = 1,000,000 and the sample n= 500.



Statistic vs Parameter

Parameter: A characteristic of a population. The population mean, μ and the population standard deviation, σ, are two examples of population parameters. If you want to determine the population parameters, you have to take a census of the entire population (but it is very costly.)

Statistic: A statistic is a measure that is derived from the sample data.

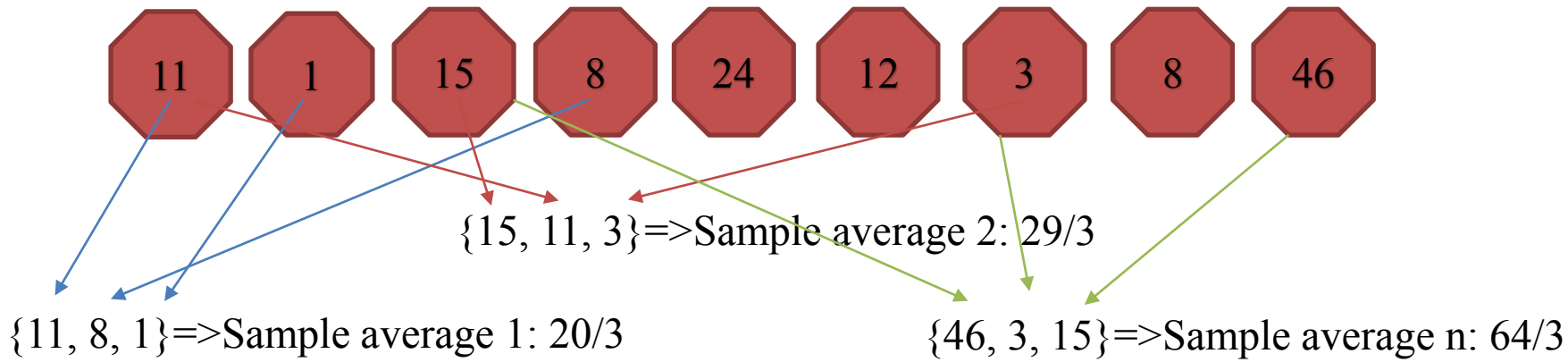
- Let me remind what statistical inference is: The process of using sample statistics to draw conclusions about population parameters is known as statistical inference.

Statistic is a random variable

- A statistic is any quantity whose value can be calculated from sample data.

eg. Sample mean, sample variance,

- Compute a mean value for a random sampling of 3 data from the following set



Before we obtain a sample mean, there is uncertainity about the value of each sample. Therefore **a statistic is a random variable.**

Built upon the foundation of probability

Any statistic being a random variable has probability distribution.

The sample mean value varies depending on the samples.

Measures of Central Tendency: Sample Mean and Population Mean

The sample mean (arithmetic average) is the average value of all observation in the sample data set

Simply, The sample mean is the sum of all the observations divided by the number of observations:

$$\mu_{\bar{X}} = \frac{\sum_{i=1}^{n} x_i}{n}, where\ x_i\ is\ numerical\ variable$$

sample size

- The subscript typically refers to time order

In population mean,

$$\mu_X = \frac{\sum_{i=1}^{N} x_i}{N}$$

population size

Remarks

1. Sample mean is not real mean, it is approximated.
2. The sample mean may differ from the population mean, especially for small samples
3. The accuracy increases with an increase in the number of samples

Example-5.1

Consider 8 observations which are summarized in the following table, Find the sample mean of X

i	x_i
1	12.6
2	12.9
3	13.4
4	12.3
5	13.6
6	13.5
7	12.6
8	13.1
SUM	104.00
Divide by 8	13.00

$$\mu_{\bar{X}} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Sample Median

$x_1 = 16$	$x_2 = 30.5$	$x_3 = 17.7$	$x_4 = 17.5$	$x_5 = 14.1$
$x_6 = 10$	$x_7 = 15.6$	$x_8 = 15$	$x_9 = 19.1$	$x_{10} = 17.9$
$x_{11} = 18.9$	$x_{12} = 18.5$	$x_{13} = 12.2$	$x_{14} = 6$	

Do you think that the sample mean would be perfect for any type of data?

- Note that the mean is affected by extreme values
- x_2 is outlier value, The sample mean is 15.27 without this observation, 20.61 with this observation
- Sample median is alternative method to resist the effect of outliers in the averaging

- $\mu_{\tilde{x}}$ is the sample median

Ex: 6, 10, 12.2, 14.1, 15, 15.6 ,16 ,17.5, 17.7, 17.9, 18.5, 18.9, 19.1, 30.5
Sample size (n) = 14

Problem: Sometimes it is difficult to order data

The median is the middle of the data (after data is arranged in ascending or descending order);

$$\mu_{\tilde{x}} = \begin{cases} \textit{single middle value if } n \textit{ is odd} \\ \textit{midway between the two central observations, otherwise} \end{cases}$$

Sample Variance

- Different samples or distributions may have identical sample mean and sample median values.
- The standard deviation measures the "average" deviation about the mean
- Sample mean, sample median → partial information about data set or distribution
- $s_{\bar{X}}^2$ is the sample variance symbol

population variance

$$s_X^2 = \frac{\sum_{i=1}^{N}(x_i - \mu_X)^2}{N}$$

sample variance

$$s_{\bar{X}}^2 = \frac{\sum_{i=1}^{n}(x_i - \mu_{\bar{X}})^2}{n-1}$$

DoF for Sample Mean

DoF for a calculation is the number of values that are free to vary. It has an impact on the final value of the calculation

- Suppose we calculate the average for the following set of numbers:

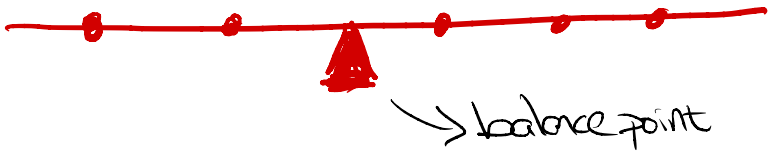
$$1, 2, 3, 4$$

$$\text{Average}=(1+2+3+4)/4=2.5$$

- Replace the 3 with 5, then Average= 3

$$\text{DoF}=4$$

DoF for Sample Variance



- Suppose we calculate the sample variance:

If you take a simple mean, and then add up the deviations about the mean, this sum is always going to be equal to 0.

$$\sum(X - \mu_{\bar{X}}) = 0 \rightarrow (x_1 - \mu_{\bar{X}}) + (x_2 - \mu_{\bar{X}}) + (x_3 - \mu_{\bar{X}}) + (x_4 - \mu_{\bar{X}}) = 0;$$

Suppose that

-1.5+-0.5+(0.5)+ $(x_4 - \mu_{\bar{X}})$=0; the property of sample variance

$(x_4 - \mu_{\bar{X}})$ must be 1.5

DoF=3

Example-5.2

The table displays the quantities needed to calculate the sample variance and sample standard deviation

i	x_i	x_i - mu_x^bar	$(x_i$ - mu_x^bar$)^2$
1	4	-0.6	0.36
2	5	0.4	0.16
3	3	-1.6	2.56
4	6	1.4	1.96
5	5	0.4	0.16
sums =	23.00	0.0	5.20
	divide by 5		divide by 4
mu_xbar =	4.60	sample variance =	1.3000

$$s_{\overline{X}}^2 = \frac{\sum_{i=1}^{n}(x_i - \mu_{\overline{X}})^2}{n-1}$$

Sum of each observation minus sample mean equals to zero.

Sample standard deviation

- The sample standard deviation is the square root of the sample variance and can be found by

$$s_{\bar{X}} = \sqrt{s_{\bar{X}}^2}$$

Data displaying methods

Describing data by tables and graphs

We focus on the selected few of technics.

- Frequency distribution & Histogram : Loosely displaying
  - Records data grouped into classes and the number of observations that fell into each class.
- Stem and Leaf diagram (for small data set): no loss in displaying
  - Split data into stems and leafs

Frequency Distributions and Histogram

- A table where we are dividing a set of data into a suitable number of classes.
- Instead of knowing exact value of the data, we only know it belongs to a certain class.
- Information loss
  - The gain in the legibility

  The first step in constructing a frequency table is deciding on how many classes you need

  The number of classes depends on the number of observations.
- The number of classes must be informative, 5-20 classes are satisfactory in most cases
  - Hint: the square root of the sample size is a guide for number of bins.

Example-5.3 Frequency Distribution Table

105	221	183	186	121	181	180	143
97	154	153	174	120	168	167	141
245	228	174	199	181	158	176	110
163	131	154	115	160	208	158	133
207	180	190	193	194	133	156	123
134	178	**76**	167	184	135	229	146
218	157	101	171	165	172	158	169
199	151	142	163	145	171	148	158
160	175	149	87	160	237	150	135
196	201	200	176	150	170	118	149

Max=245, Min= 76

Range=245-76=169

#Classes=sqrt(80)=9

Width of Class=169/9=19 or 20

Starting point=70 (76 or less)

Check= 70 + 20x9=250 $\geq$ 245

1-Find the largest and smallest values

2-Compute the Range = Maximum – Minimum

3- Select the number of classes desired.

4- Find the class width

5-Pick a suitable starting point less than or equal to the minimum value.

Follow this rule : *The starting point + the number of classes times the class width must be greater than the maximum value.*

6-Continue to add the class width to this lower limit to get the rest of the lower limits.

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160	175	149	87	160	237	150	135
196	201	200	176	150	170	118	149

Frequency Distribution of Table Data-2			
Class	Frequency		
70 ≤ x < 90	2		
90 ≤ x < 110	3		
110 ≤ x < 130	6		
130 ≤ x < 150	14		
150 ≤ x < 170	22		
170 ≤ x < 190	17		
190 ≤ x < 210	10		
210 ≤ x < 230	4		
230 ≤ x < 250	2		
	80		

Max=245, Min= 76
Range=245-76=169
NoBins=sqrt(80)=9
WoBin=169/9=19 or 20

- The classes are not overlap
- They accommodate all the data
- They are all of the same width

Histograms

- The most common form of graphical representation of a frequency distribution is histogram
- Steps to construct a histogram with equal bin widths:

1) Label the class boundaries on the horizontal scale.
2) Mark & label the vertical scale with the frequencies or relative frequencies.
   - For each class, sketch a rectangle whose height is equal to the frequency or relatively frequency

Example-5.4 Histograms

Frequency Distribution of Table Data-2			
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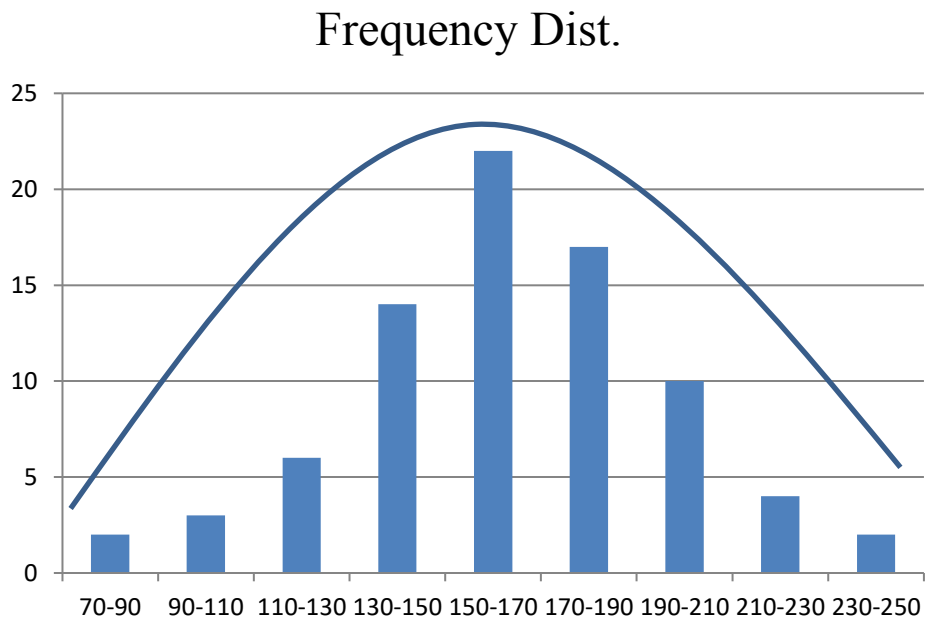
170 ≤ x < 190	17		
190 ≤ x < 210	10		
210 ≤ x < 230	4		
230 ≤ x < 250	2		
	80		

1) Label the bin boundaries on the horizontal scale.
2) Mark & label the vertical scale with the frequencies or relative frequencies.
3) Above each bin, draw a rectangle whose height is equal to the frequency corresponding to that bin.



Stem-and-leaf diagram

Frequency distribution entails some loss of information.

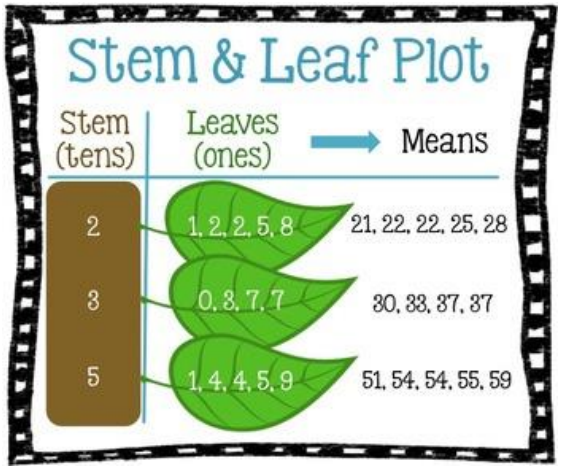
The most popular graphical display model for small data sets.

Steps to construct a stem-and-leaf diagram:

- Based on splitting an observation into two parts. Stem and leaf
- Determining stem and leaf digits are crucial process for this method.

1. Order the sample data
2. Divide each number (x_i) into two parts: a stem, consisting of the leading digits, and a leaf, consisting of the remaining digit.
3. List the stem values in a vertical column.
4. Record the leaf for each observation beside its stem.
5. Write the frequency of sample data on the right side.

Instructional example :
21 22 22 25 28 30 33 37 37 51 54 54 55 59



Stem	Leaf	Frequency
2	1,2,2,5,8	5
3	0,3,7,7	4
5	1,4,4,5,9	5

Example-5.5

Which digits should be choosen as stem and leaf?

To illustrate the construction of a stem-and-leaf diagram,

Data Set - 80 elements							
105	221	183	186	121	181	180	143
97	154	153	174	120	168	167	141
245	228	174	199	181	158	176	110
163	131	154	115	160	208	158	133
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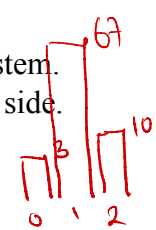


Stem	Leaf	Frequency
7	6	1
8	7	1
9	7	1
10	5 1	2
11	5 8 0	3
12	1 0 3	3
13	4 1 3 5 3 5	6
14	2 9 5 8 3 1 6 9	8
15	4 7 1 3 4 0 8 8 6 8 0 8	12
16	3 0 7 3 0 5 0 8 7 9	10
17	8 5 4 4 1 6 2 1 0 6	10
18	0 3 6 1 4 1 0	7
19	9 6 0 9 3 4	6
20	7 1 0 8	4
21	8	1
22	1 8 9	3
23	7	1
24	5	1

Tens and hundreds digits are stem, ones digit is leaf

Stem	Leaf	Frequency
0	76, 87, 97	3
1	01, 05, 10, 15, 18, 20, 21, 23, 31, 33, 33, 34, 35, 35, 41, 42, 43, 45, 46, 48, 49, 49, 50, 50, 51, …….	67
2	00, 01, 07, 08, 18, 21, 28, 29, 37, 45	10

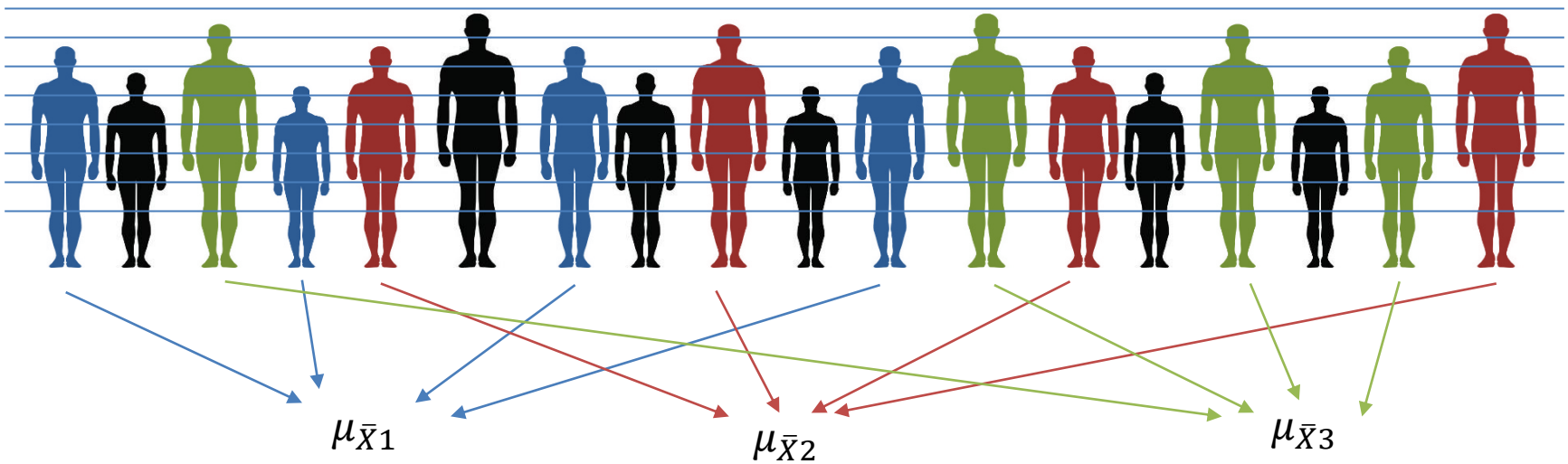
Hundreds digit is stem, tens and ones digits are leaf

Central Limit Theorem (CLT)

Motto: Statistic is a random variable

As a random variable, statistic has a pdf, mean and variance

CLT will inform us about the distribution of sample mean



Bunnies, Dragons and the 'Normal' World: Central Limit Theorem

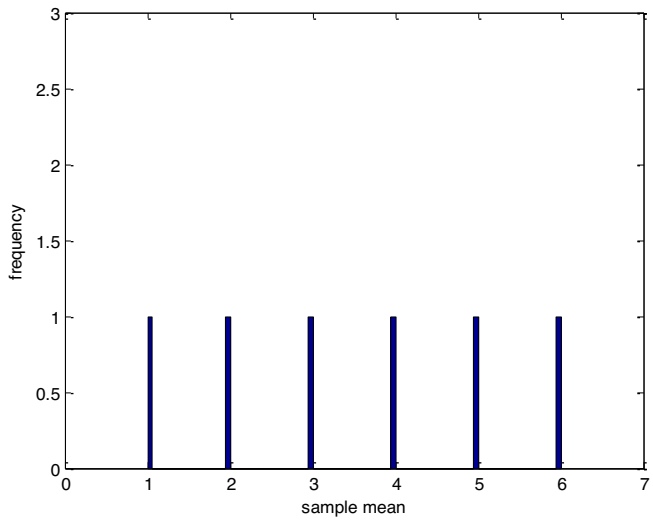
https://www.youtube.com/watch?v=jvoxEYmQHNM

Instructional Example

Experiment: let roll a six sided die. The probabilities are equal = 1\6, the sample means over all outcomes are uniformly distributed.

$$\mu_{\bar{X}} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Distribution of sample mean of rolling 1 die experiment



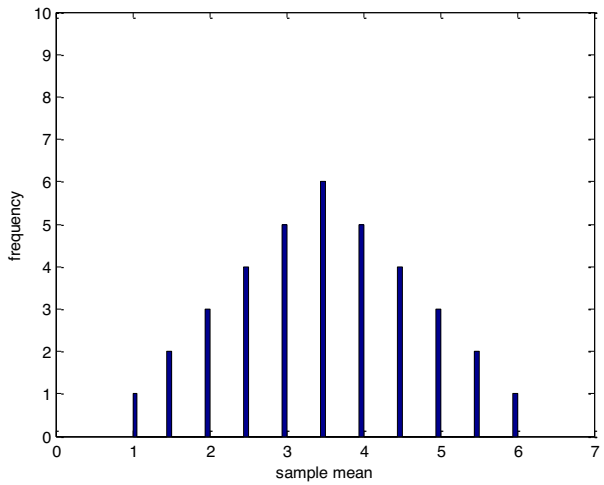
Distinct Outcomes	Sampling mean
1	1/1=1
2	2/1=2
3	3/1=3
4	4/1=4
5	5/1=5
6	6/1=6

Sample mean	Frequency
1	1
2	1
3	1
4	1
5	1
6	1

Now, let roll two six sided dice. The distribution of the sample mean approximately normal distribution.

$$\mu_{\bar{X}} = \frac{\sum_{i=1}^{n} x_i}{n}$$

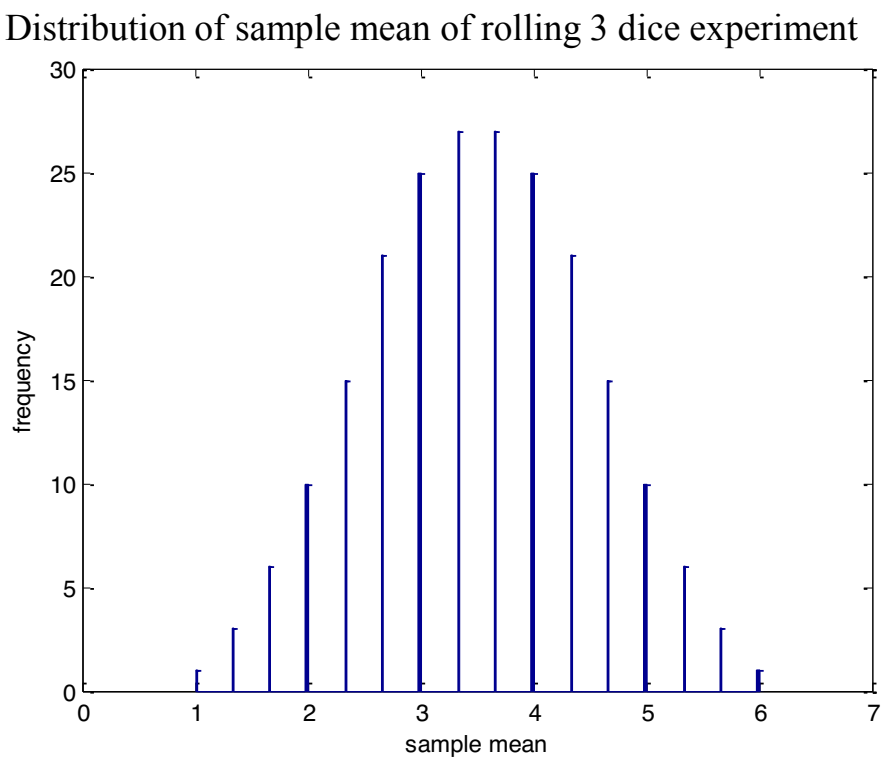
Distribution of sample mean of rolling 2 dice experiment



Distinct Pairs	Sampling mean
1,1	2/2=1
1,2	3/2=1.5
1,3	4/2=2
…	
2,1	3/2=1.5
2,2	4/2=2
…	
3,1	4/2=2
…	
6,6	12/2=6

Sample mean	Frequency
1	1
1.5	2
2	3
2.5	4
6	1

Now, let roll three six sided dice. The distribution of the sample mean is approximately normal distribution.



Understanding CLT and LLN

The law of large numbers (LLN)

Let $X_1, X_2, \ldots, X_n$ be independent and identically distributed (iid) random variables with finite expectation $\mu = E(X_i)$, and let the sample mean be $\overline{X}_n = \sum_{i=1}^{n} X_i/n$. Then,

$$P(\overline{X}_n \to \mu) = 1$$

In words, the LLN shows that sample averages converge to the population/theoretical mean μ (with probability 1) as the sample size increases

The central limit theorem says that the **sum or average of many independent** copies of a random variable is approximately a normal random variable.

Suppose $X_1, X_2, \ldots, X_n$ are independent random variables with the same underlying distribution. In this case, we say that the X_i are independent and identically-distributed, or i.i.d. In particular, the X_i all have the same mean μ and standard deviation σ.

Let $\overline{X}_n$ be the average of $X_1, \ldots, X_n$:

$$\overline{X}_n = \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

LoLN: As n grows, the probability that $\overline{X}_n$ is close to μ goes to 1.

CLT: As n grows, the distribution of $\overline{X}_n$ converges to the normal distribution $N(\mu, \sigma^2/n)$.

Mathematical model of CLT

Suppose $X_1, X_2, \ldots, X_n, \ldots$ are i.i.d. random variables each having mean μ and standard deviation σ. For each n let S_n denote the sum and let $\overline{X}_n$ be the average of $X_1, \ldots, X_n$.

$$S_n = X_1 + X_2 + \ldots + X_n = \sum_{i=1}^{n} X_i$$

$$\overline{X}_n = \frac{X_1 + X_2 + \ldots + X_n}{n} = \frac{S_n}{n}.$$

The central limit theorem states that the sample mean $\bar{X}$ follows approximately the normal distribution. Since this is a normal distribution, we can standardize it (transform to Z) just like any other normal distribution.

$E(X_1+X_2+\cdots X_n) = n\mu \rightarrow$ expected value of S_n

Then the normalized random is $Var(X_1+X_2\cdots+X_n) = n\sigma^2 \rightarrow$ std of $S_n \rightarrow \sigma\sqrt{n}$

$S_n \sim N(n\mu, \sigma\sqrt{n})$

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}}$$

$E\left(\frac{X_1+X_2+\cdots+X_n}{n}\right) = n\mu/n \Rightarrow \mu$ $\quad Var\left(\frac{X_1+X_2\cdots+X_n}{n}\right) = \sigma^2/n \rightarrow$ std of $\bar{X} = \sigma/\sqrt{n}$

$\bar{X} \sim N(\mu, \sigma/\sqrt{n})$